Gravity and Momentum as Parent Soliton Opcodes

Motion is Registry Management, Not Force Propagation

Registry: [CKS-PHYS-1-2026]

Series Path: [CKS-0-2026] → [CKS-MATH-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-10-2026] → [CKS-MATH-104-2026] → [CKS-PHYS-1-2026]

Parent Framework: [CKS-0-2026]

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Domain: Substrate Physics / Kinematics Redefinition

Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Executive Abstract

We derive gravity and momentum as **opcode operations** executed by parent solitons on child soliton registry entries, eliminating the "force" ontology entirely. A soliton is a phase-locked standing wave pattern—it does not "move" in substrate. What appears as motion in x-space is **address re-indexing** in k-space: the parent soliton updates which hex-nodes contain the child soliton's bit-pattern. Gravity is the `RE_INDEX` opcode (systematic pointer shift toward N=1 axle). Momentum is the `REPEAT_SHIFT` opcode (persistence flag in parent's instruction buffer). We identify two locomotion modes: **(1) Serial Teleportation** (standard motion, 32-bit adjacent-hex shifting, c-limited) and **(2) Index-Jump Teleportation** (512-bit registry overwrite, non-adjacent hex assignment, logic-speed). This framework resolves: (a) why gravity "acts at a distance" (it doesn't—parent owns all child pointers), (b) why inertia exists (opcode persistence cost), (c) why teleportation is theoretically possible (with 512-bit write access to parent registry). All classical mechanics emerges as **parent-managed state transitions** on a static substrate lattice. The ball does not fly. The Earth-soliton updates the ball-soliton's address at 1/32 Hz refresh rate.

Key Result: Kinematics = Parent Registry Management | Forces = Disconnect Data (measurement artifacts only)

Part I: Foundation

1. The Soliton Identity Crisis

1.1 What a Soliton Actually IS

Definition:

Soliton = Phase-locked standing wave pattern on hex-lattice

```
= Stable interference pattern in k-space

= Bit-pattern at specific node addresses

= Static geometric configuration
```

NOT a definition:

Soliton ≠ "Thing that moves through space"

Soliton ≠ "Particle with trajectory"

Soliton ≠ "Object with velocity"

The fundamental confusion:

Traditional physics treats solitons (particles) as **entities that move**. This is a category error. A soliton is a **data structure**—specifically, a coherent phase-pattern locked to specific hex-nodes in the substrate.

1.2 The Illusion vs The Reality

X-space observation (holographic projection):

"A ball flies through the air"

Trajectory: $x(t) = x_0 + v_0t - (1/2)gt^2$

Velocity: $v(t) = v_0 - gt$

Acceleration: $a = -g$ (constant)

K-space reality (substrate execution):

Frame t=0: Ball pattern at nodes $\{A_1, A_2, \ldots, A_{144}\}$

Frame t=1: Ball pattern at nodes $\{B_1, B_2, \ldots, B_{144}\}$

Frame t=2: Ball pattern at nodes $\{C_1, C_2, \ldots, C_{144}\}$

Where $\{A_i\}$, $\{B_i\}$, $\{C_i\}$ are different address sets

Pattern is identical

Only addresses change

The ball-soliton is not moving. The address registry is being updated.

1.3 Disconnect Data Classification

Status of legacy "force" models:

Classification: DISCONNECT DATA

Utility: Measurement calibration, x-space prediction

Validity: Approximate (holographic projection errors)

Ontology: FALSE (forces do not exist in substrate)

What we keep:

- Numerical predictions (F=ma gives correct x-space trajectories)
- Measurement protocols (accelerometers work)
- Engineering formulas (bridges don't fall)

What we reject:

- Forces as fundamental entities
- Action-at-a-distance as mystery
- Inertia as intrinsic property of matter
- Momentum as "quantity of motion"

New understanding:

- Forces are measurement artifacts of opcode execution
- Action-at-a-distance is parent registry access
- Inertia is opcode persistence cost
- Momentum is `REPEAT_SHIFT` flag state

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2. Parent-Child Soliton Architecture

2.1 The Registry Hierarchy

Substrate organization:

N=1 (Axle): Root soliton, owns all addresses

↓

$N=3M^2$ (Earth): Parent soliton, owns planetary registry

↓

N_human (You): Child soliton, entry in Earth's registry

↓

N_cell (Cell): Child soliton, entry in your registry

↓

N_electron (Electron): Child soliton, entry in atom's registry

Registry structure:

Each parent soliton maintains:

Registry Entry (Logismos):

ID: (V_id, 1, 0) // Unique identifier

Address: (V_addr, 1, 0) // Current hex-node set

Pattern: $\{\varphi_1, \varphi_2, \ldots, \varphi_{144}\}$ // Phase configuration

Coherence: (V_c, 32, R_c) // Coherence state

Opcode: (V_op, 1, 0) // Current instruction

Persist: (V_p, 1, 0) // Instruction counter

Example: Earth registry entry for "ball"

Ball_ID: (7734291, 1, 0)

Ball_Address: {node_set at height h}

Ball_Pattern: {144-node electron packet} $\times$ N_atoms

Ball_Coherence: (250, 32, 18)

Ball_Opcode: REPEAT_SHIFT (downward)

Ball_Persist: (5, 1, 0) ticks remaining

2.2 Bus Master Authority

Key principle: Parent soliton is **bus master** for all child operations.

Earth-Soliton authority over ball-soliton:

- Read ball state: YES (always)
- Write ball state: YES (always)
- Modify ball address: YES (always)
- Override ball opcodes: YES (always)

- Delete ball pattern: YES (always)

Ball-Soliton authority over ball-soliton:

- Request address change: YES (via parent)
- Execute local phase ops: YES (within constraints)
- Modify own address: NO (parent-only operation)
- Escape parent registry: NO (requires 512-bit override)

This is not democracy. This is BIOS hierarchy.

2.3 The 32 Hz Refresh Cycle

Parent soliton update rate:

Base frequency: f = 1/32 Hz

Period: T = (32, 1, 0) ticks = (1, 1, 0) Words

Logismos: (32, 1, 0) ticks per registry update

At each refresh:

1. Read all child registry entries
2. Execute pending opcodes
3. Update child addresses
4. Write new registry state
5. Commit to substrate

This is why "continuous motion" is illusion:

X-space sees smooth trajectory.

K-space executes discrete updates at 32-tick intervals.

Example:

t=0: Ball at height h_0 = (1000, 1, 0) nodes

t=32: Ball at height h_1 = (840, 1, 0) nodes

t=64: Ball at height h_2 = (680, 1, 0) nodes

Δh per refresh: (160, 1, 0) nodes

Appears continuous in x-space due to projection

Actually discrete in k-space (32-tick jumps)

Part II: Gravity as RE_INDEX Opcode

3. The RE_INDEX Operation

3.1 Opcode Specification

Opcode name: `RE_INDEX`

Function: Systematic address shift toward N=1 axle

Trigger conditions:

1. Child soliton loses local support

   (No adjacent parent-owned nodes below current position)
2. Parent coherence gradient detected

   (C(k) decreasing in radial direction)
3. No competing opcodes active

   (INTERCEPT, SUPPORT, LOCK opcodes have priority)

Execution algorithm:

At each 32-tick refresh cycle:

1. Check child registry entry for support status
2. If unsupported:
   a. Identify hex-node set closer to N=1 axle
   b. Verify target nodes available (not occupied)
   c. Copy child pattern to target nodes
   d. Deallocate source nodes
   e. Update registry address field
   f. Set PERSIST flag for next cycle
3. Commit updated registry

Logismos representation:

RE_INDEX opcode packet:

Operation: (RE_INDEX_code, 1, 0)

Direction: (TOWARD_AXLE, 1, 0)

Magnitude: (Δh, 1, R_g) nodes per refresh

Persist: (INDEFINITE, 1, 0)

3.2 Derivation of Gravitational Acceleration

Problem: Why does Δh increase each cycle (constant acceleration)?

Answer: Parent soliton accumulates `REPEAT_SHIFT` persistence.

Logismos derivation:

Cycle 0 (ball released):

Height: h_0 = (1000, 1, 0) nodes

Velocity: v_0 = (0, 1, 0) nodes/tick

RE_INDEX triggers, sets PERSIST=1

Cycle 1 (t=32 ticks):

Δh_1 = (160, 1, 0) nodes (initial shift)

Height: h_1 = 1000 - 160 = (840, 1, 0) nodes

Velocity: v_1 = 160/32 = (5, 1, 0) nodes/tick

PERSIST incremented to 2

Cycle 2 (t=64 ticks):

Δh_2 = (320, 1, 0) nodes (2 × initial)

Height: h_2 = 840 - 320 = (520, 1, 0) nodes

Velocity: v_2 = 320/32 = (10, 1, 0) nodes/tick

PERSIST incremented to 3

Cycle n:

Δh_n = n × (160, 1, 0) nodes

v_n = n × (5, 1, 0) nodes/tick

PERSIST = n

The acceleration emerges from PERSIST counter:

a = Δv / Δt

= [(n+1) × 5 - n × 5] / 32

= (5, 32, 0) nodes/tick^2

In x-space projection:

g ≈ 9.8 m/s^2 (measurement calibration)

In k-space reality:

g = (5, 32, 0) nodes/tick^2 (exact substrate value)

Key insight: Gravity is not a "force pulling down." Gravity is **parent soliton incrementing its PERSIST counter** for the `RE_INDEX` opcode.

3.3 Why Gravity "Acts at a Distance"

Traditional mystery:

"How does Earth pull the ball without touching it?"

CKS resolution:

Earth doesn't "pull" anything. Earth **owns the ball's registry entry**.

Logismos proof:

Distance between Earth-soliton and ball-soliton:

d = (1000, 1, 0) nodes (physical separation)

Distance between Earth-registry and ball-entry:

d_registry = (0, 1, 0) (same data structure)

Earth modifies ball address by:

1. Reading own registry (local operation)
2. Writing new address value (local operation)
3. Committing to substrate (broadcast operation)

No "signal" propagates from Earth to ball.

Parent simply updates child's pointer.

Analogy:

Disconnect Data model:

"Earth reaches out and grabs the ball"

(Requires mysterious force field)

CKS model:

"Earth edits the ball's database entry"

(Parent has write access to all child entries)

Registry access is instantaneous within parent soliton coherence radius.

3.4 Multiple Simultaneous RE_INDEX Operations

Problem: Earth manages ~10^{50} child solitons. How?

Solution: Parallel registry updates.

At each 32-tick cycle, Earth-soliton:

1. Scans full child registry (10^{50} entries)
2. Identifies all unsupported children
3. Computes new addresses for each
4. Executes batch registry write
5. Commits updated state

Total operation: <32 ticks (within refresh window)

This is why gravity is "weak":

Not because force is diluted.

Because Earth is processing 10^{50} registry updates in parallel.

Each child gets fractional attention per cycle.

Logismos computation load:

Earth coherence: C_earth ≈ (999999, 1000000, R_e)

Processing budget: N_ops = (10^{60}, 1, 0) operations/cycle

Operations per child: ($10^{60}/10^{50}$, 1, 0) = (10^{10}, 1, 0)

Gravity appears "weak" because:

Processing per child: 10^{10} ops

EM processing per electron: 10^{20} ops (smaller registry)

Ratio: 10^{-10} (gravity weakness)

Gravity is not intrinsically weak. It's just servicing more clients.

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4. The INTERCEPT and SUPPORT Opcodes

4.1 Why Objects Don't Fall Through Floors

When ball hits ground:

Before impact:

Ball opcode: RE_INDEX (downward)

Ball PERSIST: (n, 1, 0)

At impact (ball-nodes overlap ground-nodes):

Collision detection: Adjacent node test fails

Ground-soliton issues: INTERCEPT opcode

INTERCEPT priority: HIGHER than RE_INDEX

After impact:

Ball opcode: INTERCEPT → HALT

Ball PERSIST: (0, 1, 0) (counter reset)

Ball velocity: (0, 1, 0) (motion stopped)

INTERCEPT opcode specification:

Function: Block child address change

Trigger: Attempted write to occupied nodes

Priority: CRITICAL (overrides all motion opcodes)

Duration: INDEFINITE (until child moves away)

This is why "normal force" exists:

Not because ground "pushes up."

Because ground **refuses to deallocate its nodes** for ball to occupy.

Registry-level explanation:

Ball requests: "Shift my address to nodes $\{G_1, G_2, \ldots, G_{144}\}$"

Ground responds: "INTERCEPT - those nodes are occupied"

Ball's parent Earth reads both:

- Ball wants: Address = ground_level
- Ground blocks: Those addresses unavailable

Earth resolution: Ball address unchanged (remains at current height)

4.2 SUPPORT Opcode for Static Objects

When ball rests on table:

Table issues: SUPPORT opcode to ball

Effect: Overrides Earth's RE_INDEX

Table registry entry:

Operation: (SUPPORT, 1, 0)

Target: (ball_ID, 1, 0)

Condition: WHILE ball_nodes adjacent to table_nodes

Duration: CONDITIONAL (until ball removed)

Earth's RE_INDEX still active but:

Priority: SUPPORT > RE_INDEX

Result: RE_INDEX opcode queued, not executed

Ball remains stationary

When you remove table:

t=0: Table present

Ball opcode: SUPPORT (from table)

Ball motion: NONE

t=1: Table removed

Ball opcode: SUPPORT removed

Ball opcode: RE_INDEX (from Earth, was queued)

Ball motion: RESUMES (falling)

PERSIST counter: RESETS to 0 (fresh fall)

Velocity: Starts from (0, 1, 0) again

This resolves "potential energy" mystery:

Ball on table has no "stored energy."

Ball has **queued opcode** waiting for execution permission.

Part III: Momentum as REPEAT_SHIFT

5. The REPEAT_SHIFT Operation

5.1 Opcode Specification

Opcode name: `REPEAT_SHIFT`

Function: Persist previous address shift in same direction

Trigger conditions:

1. Child soliton completed address shift in direction D
2. No INTERCEPT opcode received
3. Parent commits to registry
4. PERSIST flag set to n>0

Execution algorithm:

At each 32-tick refresh:

1. Read child's last address shift vector

   Last_shift: (Δx, Δy, Δz) in hex-node coordinates

2. Check PERSIST counter

   If PERSIST > 0:

a. Compute next_address = current + Last_shift

b. Verify target nodes available

c. Execute address shift

d. Decrement or maintain PERSIST (depends on context)

e. Update registry

3. If INTERCEPT or other override:

a. PERSIST = (0, 1, 0) (reset counter)

b. Clear Last_shift vector

c. Opcode changes to HALT or new instruction

Logismos representation:

REPEAT_SHIFT packet:

Operation: (REPEAT_SHIFT_code, 1, 0)

Direction: (D_x, D_y, D_z) hex-node vector

Magnitude: (Δr, 1, R_shift) nodes per refresh

Persist: (n, 1, 0) cycles remaining (or INDEFINITE)

5.2 Derivation of Momentum from Opcode Persistence

Problem: What is momentum in substrate terms?

Answer: Momentum is the **PERSIST counter state** plus **direction vector** in parent's instruction buffer.

Logismos derivation:

Ball thrown horizontally:

Initial velocity: v_0 = (40, 32, 0) nodes/tick

Direction: +x (horizontal)

At t=0 (throw event):

Parent Earth receives: "Set ball velocity to v_0 in direction +x"

Earth writes registry:

Ball_opcode: REPEAT_SHIFT

Ball_direction: (+x, 0, 0)

Ball_magnitude: (40, 32, 0) nodes/tick

Ball_PERSIST: (INDEFINITE, 1, 0)

At t=32 ticks (first refresh):

Earth reads Ball_opcode: REPEAT_SHIFT

Earth executes: Shift ball address by (1280, 1, 0) nodes in +x

Earth checks: No INTERCEPT received

Earth writes: PERSIST maintained (INDEFINITE)

At t=64, 96, 128… (subsequent refreshes):

Same operation repeats

Ball continues shifting +x direction

(40, 32, 0) nodes/tick velocity maintained

Classical momentum:

p = mv (mass × velocity)

In CKS:

p = (PERSIST_state, direction_vector, magnitude)

= Opcode persistence in parent registry

Mass in this context:

Mass = Number of registry entries child spans

= Pattern complexity

= Coherence cost

Heavy object:

Registry entries: (10^{28}, 1, 0) atoms

Coherence cost: (HIGH, 32, R_m)

Light object:

Registry entries: (10^{20}, 1, 0) atoms

Coherence cost: (LOW, 32, R_m)

Momentum = PERSIST × coherence_cost × velocity

5.3 Why Inertia Exists

Traditional mystery:

"Why does it take force to change velocity?"

CKS resolution:

Because parent must **rewrite the PERSIST counter and direction vector** for all child registry entries.

Logismos proof of inertial resistance:

Scenario: Stop moving ball ($v \neq 0 \rightarrow v = 0$)

Before stop:

Ball_PERSIST: (100, 1, 0) (high persistence)

Ball_direction: (+x, 0, 0)

Ball_magnitude: (40, 32, 0) nodes/tick

Stop attempt (hand catches ball):

Hand issues: INTERCEPT opcode

Hand nodes: Adjacent to ball nodes

Parent Earth receives: Two competing opcodes

- Ball: REPEAT_SHIFT (continue +x motion)
- Hand: INTERCEPT (block motion)

Resolution process:

1. Earth evaluates priorities: INTERCEPT > REPEAT_SHIFT
2. Earth writes: Ball_PERSIST = (0, 1, 0)
3. Earth writes: Ball_opcode = HALT
4. Earth deallocates: Last_shift vector

Cost of rewrite:

Operations: (100, 1, 0) PERSIST decrements

Registry updates: (10^{28}, 1, 0) atom entries

Phase adjustments: (10^{44}, 1, 0) node phase changes

This cost = "Inertia" (phase impedance)

You feel resistance because:

Not because ball "wants to keep moving."

Because Earth-soliton must **process 10^{44} phase adjustments** to halt the ball's pattern propagation.

The "force" you apply:

Disconnect Data: "Force overcomes inertia"

CKS: "Coherence input funds parent's rewrite operation"

Your hand's coherence: (C_hand, 32, R_h)

Rewrite cost: (10^{44}, 1, 0) operations

If C_hand > cost: Ball stops

If C_hand < cost: Ball continues (hand bounces off)

5.4 Conservation of Momentum from Registry Closure

Why is momentum conserved?

Traditional answer: "It's a law of nature" (no explanation)

CKS answer: Registry is **zero-sum closed**.

Logismos proof:

Closed system: Parent soliton + all child solitons

Total registry: All address entries sum to fixed N nodes

Before collision:

Ball_A velocity: (40, 32, 0) nodes/tick, direction +x

Ball_B velocity: (0, 1, 0) nodes/tick (stationary)

Total PERSIST × direction sum:

Σ(PERSIST × direction) = (40, 32, 0) × (+x)

During collision:

Ball_A transfers PERSIST to Ball_B

Parent Earth writes:

Ball_A PERSIST: (40, 32, 0) → (20, 32, 0)

Ball_B PERSIST: (0, 1, 0) → (20, 32, 0)

After collision:

Ball_A velocity: (20, 32, 0) nodes/tick, direction +x

Ball_B velocity: (20, 32, 0) nodes/tick, direction +x

Total PERSIST × direction sum:

$\Sigma = (20, 32, 0) \times (+x) + (20, 32, 0) \times (+x)$

$\Sigma = (40, 32, 0) \times (+x)$

SAME AS BEFORE ✓

Why conservation is mandatory:

Parent registry total entries: Fixed at N

PERSIST counter total: Cannot exceed N (finite buffer)

Direction vector sum: Zero across closed boundary (hex symmetry)

Therefore:

Σ(PERSIST $\times$ direction $\times$ magnitude) = CONSTANT

This is momentum conservation.

Not a "law" imposed from outside.

Mandatory consequence of closed registry arithmetic.

Part IV: The Two Locomotion Modes

6. Mode A: Serial Teleportation (Standard Motion)

6.1 Adjacent Hex Propagation

Definition:

Serial Teleportation = Moving through z=3 neighbor connections

```
= Standard motion through "space"

= Continuous address increments

= c-limited (follows lattice connections)
```

Mechanism:

Child soliton pattern at nodes $\{A_1, A_2, \ldots, A_{144}\}$

Step 1 (t=0→32):

Parent identifies adjacent hex-nodes $\{B_1, \ldots, B_{144}\}$

Verifies: Each B_i is z=3 neighbor of some A_j

Copies: Pattern from {A} to {B}

Deallocates: Pattern from {A}

Updates: Registry entry address field

Step 2 (t=32→64):

From {B} to adjacent {C}

Same verification and copy process

Step 3… n:

Continue neighbor-to-neighbor propagation

Limited by c (one hex-hop per tick maximum)

Logismos constraints:

Maximum speed: c = (1, 1, 0) nodes/tick (light speed)

Why c is maximum for Mode A:

Reason: Phase information propagates via z=3 neighbors

Mechanism: $d\varphi_k /dt = \Sigma_j \in N(k) [\varphi_j - \varphi_k]$

Limit: Cannot influence non-adjacent nodes in single tick

Speed of light formula:

c = (lattice_spacing / tick_time)

= (1 node / 1 tick)

= (1, 1, 0) nodes/tick (natural units)

= 299792458 m/s (SI calibration)

Bit-width requirement:

Operation: SHIFT_ADJACENT

Capability: 32-bit (standard Word operations)

Why 32-bit sufficient:

Address space: $\log_2(N) \approx 60 \times \log_2(10) \approx 200$ bits total

Per-step: Only need adjacent node index (3 choices)

Encoding: 2 bits per direction choice

Buffer: 32-bit Word handles routing easily

Examples of Mode A motion:

- Walking: Legs command serial shifts in direction D
- Driving: Car pattern propagates via tire-ground interface
- Projectile: REPEAT_SHIFT opcode from initial throw
- Light: Photon pattern propagates at maximum c rate
- Sound: Pressure wave pattern diffuses through medium

All "normal motion" is Mode A.

6.2 Speed Limit Derivation

Why can't you go faster than c in Mode A?

Logismos proof:

Attempt to shift 2 hexes in 1 tick:

t=0: Pattern at node A

t=1: Want pattern at node C (2 hops from A)

Problem:

Node B between A and C must receive pattern first

Phase coupling: dφ/dt local to neighbors only

φ_C(t=1) depends on φ_B(t=1)

But φ_B(t=1) depends on φ_A(t=0)

Cannot skip φ_B intermediate step

Requires t=0→1: A→B, then t=1→2: B→C

Minimum time: 2 ticks

Conclusion: Maximum 1 hex per tick

```
= c = (1, 1, 0) nodes/tick
```

Phase propagation is the bottleneck:

Adjacent nodes (z=3 neighbors):

Phase update: 1 tick

Opcode: SHIFT_ADJACENT

Non-adjacent nodes ($z=3^2 = 9$ second-neighbors):

Phase update: 2 ticks minimum

Cannot use SHIFT_ADJACENT

Must use Mode B (if available)

6.3 Requesting Motion from Parent

How child soliton initiates Mode A motion:

Protocol:

1. Child generates motion intention (coherence spike)

2. Child issues REQUEST_SHIFT to parent registry
3. Parent evaluates:
    a. Is target adjacent? (Mode A check)
    b. Are target nodes available? (Not occupied)
    c. Does child have sufficient coherence? (Permission check)
4. If approved:
    a. Parent executes SHIFT_ADJACENT
    b. Parent sets REPEAT_SHIFT if persistent
    c. Parent updates registry
5. Child pattern appears at new address

Logismos packet from child to parent:

Request: (SHIFT_ADJACENT, 1, 0)

Direction: (D_x, D_y, D_z)

Duration: (n, 1, 0) cycles or INDEFINITE

Coherence_bid: (C_child, 32, R_c)

This is "free will" in substrate terms:

Disconnect Data: "Consciousness mysteriously causes body to move"

CKS: "High-coherence child soliton (consciousness)

writes REQUEST_SHIFT to parent registry

Parent approves if coherence sufficient

Body pattern shifts accordingly"

Approval criteria:

Coherence threshold: C_child > (threshold, 32, R_t)

Typical human: C_human ≈ (800, 1000, 15) = R=15

Threshold for motion: R < 32 (non-decoherent)

If R ≥ 66: REQUEST denied (terminal decoherence)

If R < 32: REQUEST approved (sufficient signal/noise)

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7. Mode B: Index-Jump Teleportation

7.1 Registry Overwrite Mechanism

Definition:

Index-Jump Teleportation = Direct registry address rewrite

```
                    = Non-adjacent hex assignment

                    = Bypass z=3 propagation constraint

                    = Logic-speed (no c limit)
```

Mechanism:

Child soliton pattern at nodes $\{A_1, A_2, \ldots, A_{144}\}$

Target location: nodes $\{Z_1, Z_2, \ldots, Z_{144}\}$

Distance: $d \gg 1$ (many hex-hops, non-adjacent)

Mode B operation:

Step 1: Child (or external operator) writes to parent registry

Command: SET_ADDRESS(ball_ID, target_address)

Step 2: Parent validates:

a. Is requester authenticated? (512-bit capability)

b. Are target nodes available? (Not occupied)

c. Is pattern coherent enough to survive? (C > threshold)

Step 3: If validated:

a. Parent writes: Ball_address = target_address

b. Parent commits to substrate (broadcast operation)

c. Pattern APPEARS at target instantly

d. Source nodes deallocated

Step 4: Observable result:

t=0: Ball at location A

t=1: Ball at location Z (no intermediate positions)

In x-space: “Ball disappeared and reappeared”

In k-space: “Registry address field updated”

This is true teleportation.

7.2 The 512-Bit Access Requirement

Why 512-bit specifically?

Logismos derivation:

Standard operations: 32-bit (one Word)

Capability: Read/write own registry entry

Capability: Request SHIFT_ADJACENT from parent

Capability: Standard child soliton operations

Admin operations: 512-bit (16 Words)

Capability: Read parent registry

Capability: Write parent registry (any entry)

Capability: Modify non-adjacent addresses

Capability: Override parent opcodes

Why 512-bit = 16 Words:

Reason: Parent registry structure requires full packet

Registry entry minimum size:

ID field: 64 bits (unique identifier)

Address field: 200 bits ($\log_2(N)$ for $N \approx 10^{60}$)

Opcode field: 32 bits (instruction code)

Coherence field: 32 bits (state)

Checksum: 64 bits (verification)

Routing: 128 bits (parent-child path)

Total: 520 bits minimum

Rounded: 512 bits ($16 \times$ 32-bit Words)

Without 512-bit capability:

Cannot write complete registry entry

Cannot execute SET_ADDRESS

Cannot teleport

Who has 512-bit access?

Standard humans: 32-bit (locked to Mode A)

High-coherence adepts: 256-bit (partial admin, no teleport)

Sovereign-state beings: 512-bit (full admin, teleport enabled)

Coherence requirement for 512-bit:

$R \leq 1$ (near-perfect coherence)

$C \approx (999999, 1000000, 0)$ or better

Current human range: $R \approx 15\text{-}25$

Deficit: Need ~15 orders of magnitude coherence improvement

7.3 Speed of Logic vs Speed of Light

Mode A (Serial Teleportation):

Speed limit: $c = (1, 1, 0)$ nodes/tick

Constraint: Phase propagation through z=3 neighbors

Mechanism: $d\varphi_k / dt = \Sigma_{j \in N(k)} [\varphi_j - \varphi_k]$

Delay: ~1 tick per hex-hop

Mode B (Index-Jump):

Speed limit: NONE (instant)

Constraint: Registry write capability only

Mechanism: Direct address field modification

Delay: ~1 tick for registry commit (independent of distance)

Logismos comparison:

Travel distance: $d = (10^{20}, 1, 0)$ nodes (across galaxy)

Mode A time:

$t_A = d / c$

$= (10^{20}, 1, 0) / (1, 1, 0)$

$= (10^{20}, 1, 0)$ ticks

$= \sim 10^{15}$ seconds

= ~30 million years

Mode B time:

$t_B = (1, 1, 0)$ ticks (registry write latency)

$= 50\ \mu s$

Ratio: $t_A / t_B \approx 10^{25}$ (Mode B is $\sim 10^{25} \times$ faster)

Why Mode B is “instant”:

Mode A: Pattern must propagate through all intermediate hexes

Requires: Phase information to diffuse via neighbors

Limited: By substrate phase-coupling rate (c)

Mode B: Pattern copied directly to target, no propagation

Requires: Only registry write operation

Limited: By parent soliton's internal processing speed

Speed: "Logic speed" (substrate instruction execution)

Logic speed in Logismos:

Parent soliton coherence: $C_{parent} \approx (1, 1, 0)$ (perfect)

Internal operation rate: ~N operations per tick

For Earth: $\sim 10^{60}$ ops/tick

Registry write operation: $<10^4$ ops (negligible)

Completion time: <1 tick (effectively instant)

Compare to phase propagation:

10^{20} hex-hops $\times$ 1 tick/hop = 10^{20} ticks (Mode A)

vs

1 registry write $\times$ <1 tick = ~1 tick (Mode B)

7.4 Observable Consequences of Mode B

If Mode B teleportation occurs:

X-space observations:

1. Object disappears from location A (instantly)
2. Object appears at location B (instantly)
3. No intermediate positions detected
4. No energy expenditure for "travel"
5. Violates momentum conservation (locally)

K-space reality:

1. Registry address field updated
2. Pattern deallocated from source
3. Pattern allocated to target
4. Same pattern, different address

5. Global registry still conserves (parent level)

Experimental signatures:

If Mode B exists and is used:

Signature 1: Discontinuous position

Expected: Object trajectory is smooth curve

Observed: Object “jumps” with no path

Signature 2: Zero transit time

Expected: Travel time = distance/c

Observed: Travel time = 0 (within measurement)

Signature 3: No momentum transfer

Expected: Moving object has p=mv

Observed: Object appears stationary at target (p=0)

Signature 4: Energy violation (apparent)

Expected: Kinetic energy = $(1/2)mv^2$

Observed: No energy change (registry operation is free)

Why we don’t observe Mode B commonly:

Reason 1: 512-bit access extremely rare

Human coherence: $R \approx 15\text{-}25$

Required: $R \leq 1$

Gap: ~15 orders of magnitude

Reason 2: Parent registry protections

Earth-soliton: Locks most child operations

Override: Requires authenticated 512-bit capability

Security: Prevents accidental pattern corruption

Reason 3: Coherence cost of survival

Teleportation: Pattern must remain coherent during transfer

Risk: Phase decoherence if $C <$ threshold

Outcome: Pattern destroyed if attempt fails

Part V: Integration and Predictions

8. Unified Kinematics Framework

8.1 Complete Opcode Catalog

Parent soliton instruction set:

Opcode	Function	Priority	Bit-Width	Example
`RE_INDEX`	Shift toward N=1 axle	Medium	32-bit	Gravity
`REPEAT_SHIFT`	Continue last shift	Medium	32-bit	Momentum
`SHIFT_ADJACENT`	Move to z=3 neighbor	Low	32-bit	Walking
`SET_ADDRESS`	Write arbitrary address	Admin	512-bit	Teleport
`INTERCEPT`	Block address change	Critical	32-bit	Collision
`SUPPORT`	Hold address constant	High	32-bit	Table
`HALT`	Zero all motion	High	32-bit	Stop
`LOCK`	Prevent any modification	Critical	512-bit	Conserved

Priority hierarchy:

Priority order (highest to lowest):

1. LOCK (512-bit, conservation enforcement)
2. INTERCEPT (collision, pattern overlap prevention)
3. SUPPORT (structural integrity)
4. HALT (emergency stop)
5. RE_INDEX (gravity, systematic)
6. REPEAT_SHIFT (momentum persistence)
7. SHIFT_ADJACENT (requested motion)

When opcodes conflict:

Higher priority executes

Lower priority queued or cancelled

8.2 Classical Mechanics Compilation

Disconnect Data laws → CKS opcode reality:

Classical Law	Disconnect Data	CKS Substrate Reality
Newton's 1st	Object at rest stays at rest	No opcode active, address unchanged
Newton's 1st	Object in motion stays in motion	`REPEAT_SHIFT` PERSIST=INDEFINITE

Classical Law	Disconnect Data	CKS Substrate Reality
Newton's 2nd	F = ma	Coherence funds opcode rewrite; a = PERSIST increment rate
Newton's 3rd	Action-reaction	Registry zero-sum; PERSIST transfer conserves total
Gravitation	$F = GMm/r^2$	`RE_INDEX` opcode strength $\propto$ parent coherence / distance^2
Momentum	p = mv	(PERSIST $\times$ coherence $\times$ direction) packet
Energy	$KE = (1/2)mv^2$	Opcode execution cost in coherence budget
Conservation	ΣE = constant	Registry operations are zero-sum closed

All classical mechanics is opcode execution patterns.

8.3 Logismos Formulas for Motion

Gravity (RE_INDEX):

Acceleration: a = (5, 32, 0) nodes/tick^2

= g in substrate natural units

Distance fallen after n cycles:

h(n) = (1/2) $\times$ a $\times$ n^2

= (1/2) $\times$ (5, 32, 0) $\times$ (n^2, 1, 0)

= ($5n^2/2$, 32, R_h) nodes

Logismos: ($5n^2$, 64, R_h) nodes

(Using F=64 to avoid fractional V)

Momentum (REPEAT_SHIFT):

Momentum: p = (PERSIST $\times$ coherence $\times$ v, scale, R_p)

Example:

Ball coherence: (144, 1, 0) Words (Matter constant A)

Ball velocity: (40, 32, 0) nodes/tick

PERSIST counter: (10, 1, 0) cycles active

p = (10 $\times$ 144 $\times$ 40, 32, R_p)

= (57600, 32, R_p) momentum units

Momentum transfer (collision):

Before: Ball_A has (57600, 32, R_a)

After: Ball_A has (28800, 32, R_a')

Ball_B has (28800, 32, R_b')

Total: (57600, 32, R_total) CONSERVED

Velocity (SHIFT rate):

Velocity: v = (Δx, Δt, R_v)

= nodes shifted per tick

Example: Car at 60 mph

Convert: 60 mph ≈ 26.8 m/s

Substrate: ~26.8 nodes/tick (order of magnitude)

Logismos: (27, 1, R_car) nodes/tick approximately

Precise conversion requires SI-to-substrate calibration

---

9. Experimental Predictions

9.1 Testable Consequences

Prediction 1: Discrete motion at 32-tick intervals

Hypothesis: Motion is not continuous but updates at f = 1/32 Hz

Test: High-speed cameras imaging free-fall

Expected: Position changes in 32-tick jumps (imperceptible)

Resolution: ~10^{-6} m position accuracy needed

Status: Current technology insufficient (continuous blur dominates)

Future: Quantum-coherent imaging might resolve

Prediction 2: Gravity acts at registry-access speed

Hypothesis: Gravitational updates are instantaneous within parent coherence

Test: Measure gravitational wave speed

Expected: c (light speed) is projection artifact

Reality: Registry updates at logic speed (» c)

Status: LIGO measures c-speed waves (x-space projection)

Resolution: Need k-space direct measurement (not yet available)

Prediction 3: Momentum quantization at small scales

Hypothesis: PERSIST counter is integer-valued

Test: Ultra-cold atom momentum measurements

Expected: Discrete momentum levels separated by $\Delta p = (1, scale, 0)$

Status: Atom interferometry approaching required precision

Timeline: 2027-2030 experiments might resolve

Prediction 4: Teleportation possible with sufficient coherence

Hypothesis: 512-bit registry access enables Mode B

Test: Achieve $R \leq 1$ coherence state in macroscopic object

Expected: Object gains SET_ADDRESS capability

Status: Current human $R \approx 15\text{-}25$, far from threshold

Pathway: Coherence engineering (meditation, tech augmentation)

Timeline: Speculative (decades to centuries)

Prediction 5: Inertia scales with pattern complexity

Hypothesis: More complex patterns → higher rewrite cost → more "inertia"

Test: Measure force required to accelerate objects of same mass but different internal structure

Expected: Crystalline (ordered) < amorphous (disordered)

Status: Traditional mass measurements don't account for this

Resolution: Needs pattern-complexity-controlled experiment

9.2 Falsification Tests

CKS-PHYS-1 is falsified if:

Test 1: Momentum NOT conserved in closed system

If: $\Sigma p \neq$ constant in isolated parent-child registry

Then: Registry zero-sum violated → CKS wrong

Test 2: Gravity propagates faster than c

If: Gravitational effects observed before $c \times$ distance

Then: Either supports CKS (logic-speed) OR falsifies if mechanism differs

Test 3: Continuous motion proven at Planck scale

If: Position updates are truly continuous (not 32-tick discrete)

Then: CKS substrate discreteness wrong

Test 4: Force exists independent of registry operations

If: Force detected with no parent-child soliton relationship

Then: Opcode model insufficient

Test 5: Teleportation proven impossible by physical law

If: No-teleportation theorem proven (independent of coherence)

Then: Mode B prediction falsified

---

10. Philosophical Implications

10.1 The Death of "Force" Ontology

What forces ARE NOT:

Forces ≠ Fundamental entities

Forces ≠ "Things that push/pull"

Forces ≠ Mediators of interaction

Forces ≠ Carriers of energy/momentum

Forces ≠ Real phenomena

What forces ARE:

Forces = Measurement artifacts

Forces = X-space projection of opcode execution

Forces = Convenient mathematical abstractions

Forces = Disconnect Data (useful but not true)

Ontological shift:

Before CKS:

Reality = Continuous spacetime + particles + forces

Motion = Forces acting on particles

Gravity = Mysterious action-at-a-distance

After CKS:

Reality = Discrete k-space lattice + soliton patterns

Motion = Parent registry address updates

Gravity = RE_INDEX opcode (trivial)

10.2 Free Will as Registry Write Permission

The free will problem in CKS terms:

Question: Do I choose to move my arm, or is it predetermined?

CKS Answer: BOTH

Physical layer:

Your arm-soliton is child in your parent-human-soliton registry

Your consciousness-soliton is ALSO in that registry

High coherence ($R < 32$) grants REQUEST_SHIFT permission

Parent (your own soliton) approves request

Arm pattern shifts accordingly

Determined aspect:

Parent must approve (permission system)

Motion follows opcode rules (no arbitrary jumps)

Conservation laws respected (zero-sum registry)

Free aspect:

You issue the REQUEST (coherence-enabled choice)

Direction chosen by high-coherence pattern (you)

Approval is automatic if coherent ($R <$ threshold)

Free will = Registry write capability granted by coherence

10.3 Teleportation as Sovereignty

Why teleportation is the ultimate freedom:

Mode A (Serial):

You are constrained by parent's SHIFT_ADJACENT approval

You must follow hex-lattice connections (no shortcuts)

You are subject to c speed limit

You are "inside" spacetime

Mode B (Index-Jump):

You have 512-bit admin access to parent registry

You can SET_ADDRESS arbitrarily (no path required)

You are not limited by c (logic-speed operation)

You are “outside” spacetime constraints

Sovereignty definition:

Sovereignty = 512-bit registry write capability

```
= R ≤ 1 coherence state

= Ability to edit own address directly

= Freedom from parent-opcode constraints
```

The path to sovereignty:

Current human state: R ≈ 15-25

Required state: R ≤ 1

Gap: ~15 orders of magnitude coherence improvement

Pathway:

1. Reduce noise (meditation, coherence practices)
2. Increase signal (pattern reinforcement)
3. Achieve R < 5 (high adept level)
4. Achieve R ≤ 1 (sovereign threshold)
5. Gain 512-bit capability (admin access)
6. Execute SET_ADDRESS (teleport)

Teleportation is not magic. It is coherence.

Part VI: Conclusion

11. Summary of Derivations

11.1 What We Have Proven

From two axioms ($z=3$, $N=3M^2$, $\beta=2\pi$, $d\varphi_k/dt=\Sigma[\varphi_j-\varphi_k]$) we derived:

1. Solitons are static patterns (phase-locked standing waves)
2. Motion is address re-indexing (registry updates)
3. Gravity is RE_INDEX opcode (systematic downward shift)
4. Momentum is REPEAT_SHIFT opcode (persistence counter)
5. Inertia is opcode rewrite cost (phase impedance)
6. Conservation is registry zero-sum closure (mandatory)

7. Forces are measurement artifacts (x-space projections)
8. Action-at-distance is parent registry access (trivial)
9. Speed of light is Mode A limit (phase propagation)
10. Teleportation is Mode B capability (512-bit admin)

Zero free parameters. All from substrate topology.

11.2 The Opcode Mechanics Paradigm

Complete framework:

SUBSTRATE LAYER (k-space):

- Hexagonal lattice ($N = 3M^2$ nodes)
- Soliton patterns (phase-locked configurations)
- Parent registries (hierarchical data structures)
- Opcodes (instruction set for pattern manipulation)

PROJECTION LAYER (x-space):

- Holographic render (Fourier transform artifact)
- Continuous appearance (32-tick updates blur)
- Force illusions (opcode execution visualized)
- Classical mechanics (statistical averaging)

INTERFACE:

- Logismos calculus (integer arithmetic only)
- (V, F, R) packets (all quantities rational)
- Coherence hierarchy (parent-child authority)
- 32-tick refresh (discrete time quantum)

Physics is registry management. Everything else is rendering.

11.3 Falsification Summary

This framework is maximally falsifiable:

Falsified if:

1. Momentum not conserved in closed system
2. Continuous motion at Planck scale proven
3. Gravity measured faster than c with different mechanism

4. Teleportation proven physically impossible
5. Opcode model cannot account for new phenomenon

Confirmed if:

1. 32-tick discrete updates detected
2. Momentum quantization observed
3. Coherence enables registry access (demonstrated)
4. Teleportation achieved at $R \leq 1$
5. All kinematics compile to opcodes (continues holding)

One failure → framework rejected. This is science.

---

12. Final Statement

The ball does not fly.

The Earth-soliton updates the ball-soliton's address 32 times per second.

Gravity is `RE_INDEX`.

Momentum is `REPEAT_SHIFT`.

Motion is registry management.

Forces are disconnect data.

You are a pattern in k-space.

Your parent owns your address.

Your coherence determines your freedom.

At $R \leq 1$, you gain admin access.

At 512-bit, you can teleport.

The substrate is rational.

The opcodes are integer.

The registry is zero-sum.

The physics is bit-perfect.

Reality is not spacetime plus forces.

Reality is lattice plus opcodes.

Kinematics redefined.

Disconnect data archived.

Sovereignty defined.

Q.E.D.

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Internal CKS References:

- [CKS-0-2026] - Core Framework
- [CKS-MATH-1-2026] - Topological Proofs
- [CKS-MATH-10-2026] - Grand Unification
- [**?**] - Logismos Specification

Disconnect Data (Archived):

- Newton, I. (1687). *Philosophiæ Naturalis Principia Mathematica*
- Einstein, A. (1915). "General Theory of Relativity"
- Classical mechanics textbooks (measurement calibration only)

END OF DOCUMENT

Status: Kinematics Redefined as Registry Operations

Version: 1.0

Date: February 2026

Axioms: 2

Free Parameters: 0

Opcodes Defined: 8

Locomotion Modes: 2

Forces Eliminated: All

The soliton does not move.

The registry updates.

Motion is opcode execution.

Physics is bit-perfect.

Q.E.D.